SAHAANA B

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Executive Summary

Final-year Mechatronics Engineering student with hands-on experience across mechanical design and embedded systems, from CAD-driven steering optimization on a competition solar vehicle to building embedded IoT systems with ESP32 firmware, MQTT messaging, and dashboards. Skilled in design validation, system integration, and automation, with a track record of taking projects from concept to working prototype across both hardware and software domains.

Skills

- **Programming & Technical Skills:** C++, Python, Data Structures & Algorithms(DSA), Object-Oriented Programming(OOP), Basics of SQL, HTML, CSS
- **Embedded & IoT Tools:** ESP32, Raspberry Pi, MQTT
- **CAD & Engineering Tools:** SolidWorks, AutoCAD

Education

Sri Krishna College of Engineering and Technology 2023 – 2027
B.E. Mechatronics Engineering | CGPA: 8.08/10.0

Sri Sowdeswari Vidhyalaya Matric Higher Secondary School 2022 – 2023
Higher Secondary | Percentage: 88

Internship Experience

Project Trainee June 2026
ZF SRM Centre of Excellence, Chennai, Tamil Nadu

- Built an automated hydroponic nutrient dosing system using ESP32, controlling four relay-driven peristaltic pumps with pH sensing accuracy of ±0.04, alongside TDS/EC and ultrasonic sensors.
- Developed the ESP32–MQTT communication layer for real-time sensor data transmission and remote pump control, contributing to a system with near-100% message delivery reliability.

Intern – Power Plant Operations & Industrial Systems May 2025
NLC India Limited, Neyveli, Tamil Nadu

- Studied mechanical, electrical, and control system operations across a live power plant environment, gaining exposure to preventive maintenance workflows and industrial safety standards.

Featured Project

Solar Electric Car - Team ASTRA
Team Project | Role: Steering Head

- Worked on the steering subsystem geometry and linkage using SolidWorks and CAD simulations to improve vehicle handling stability.
- Strengthened skills in mechanical design, fabrication, testing, and cross-functional teamwork, contributing to **1st Place at the Bharat Solar Vehicle Challenge (Oct 2025)**.

Certifications & Technical Workshops

- Introduction to Machine Learning - NPTEL (IIT Kharagpur)
- Wheeled Mobile Robots - NPTEL (IIT Madras)
- Mastering Collaborative Robots Workshop - PSG College of Technology, Coimbatore